

Ángel Hurtado Flores — Embedded Software Engineer

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Summary

Embedded Software Development Engineer at Amazon, focused on real-time media streaming and low-level systems. Experienced in C/C++20, Embedded Linux and real-time, performance-critical systems.

At Amazon, I took technical ownership of the streaming stack for hardware prototypes approaching launch, ensuring production readiness through targeted feature development, bug fixing, and post-launch observability. As streaming PoC for various Ring devices, I coordinated closely with other teams, engineers and stakeholders to prioritize issues and deliver reliable customer-facing releases.

Experience

- Jun 25 – Present  **Amazon** — Embedded Software Development Engineer Madrid, Spain
Full-time
- Amazon Devices – Ring.** Firmware Streaming team.
- Embedded C and C++20
 - Embedded Linux and FreeRTOS
 - Streaming (RTP, SDP, GCC/TWCC, H264/H265, OPUS, BW Adaptation)
 - Observability tools (Splunk, Tableau)
 - GenAI (Kiro, Claude Code, Agentic workflows)
- Key achievements:**
- Technical owner and primary point of contact for streaming pipeline integration in new Ring hardware devices.
 - Delivered several end-to-end streaming features, from firmware implementation and debugging to post-launch observability and production monitoring.
 - Diagnosed and resolved critical customer-facing issues beyond streaming, including audio processing, PoE integration (**resulting in x3 Ethernet performance**) and cloud interaction layers.
 - Reduced development time by **40-60%** by leveraging GenAI/agent tools. **Top 1000** Kiro IDE users from all +300,000 corporate employees at Amazon.
 - Contributed to hiring process of Embedded Engineers.
- Sep 24 – Mar 25  **Safran** — Embedded Software Engineer Granada, Spain
Internship
- Embedded C
 - Embedded Linux
 - Time synchronization protocols (White Rabbit, PTP, NTP)
- Key achievements:**
- Modernized the NTP system in our devices, enhancing performance, security, and adding new functionalities.
 - Developed an automated testing system to validate the complete combination of use cases for our devices, enabling proactive detection of bugs in core functionalities.
- Mar 24 – Sep 24  **Research Assistant** — CITIC, Universidad de Granada Granada, Spain
Learnt from a researcher awarded an ERC Grant.
- Development and debugging of embedded systems
 - Control & coordination of drone swarms
 - Focus on swarm robotics

Education

- 2024 – 2025  **Master's degree in Data Science & Computer Engineering** Universidad de Granada
Specialization in Computer Engineering
Grade: 8.8 out of 10
- 2020 – 2024  **Bachelor's degree in Computer Science & Engineering** Universidad de Granada
Specialization in Computer Engineering
Grade: 8 out of 10

Skills

Languages	English (C1 Advanced), Spanish (Native), French (basic)
Soft Skills	Communication, critical thinking, leadership, teamwork, adaptability
Coding	C++, C, Python, Rust, Java
Embedded	Embedded C, Embedded Linux, Assembly (x86 & ARM), FreeRTOS, CMake
Tools	Git, Linux, Bash Scripting, Docker, clang-tools Google Cloud
AI	NPU acceleration, Edge LLMs, Embedded AI, Agentic Workflows

Certifications & Courses

Certifications

2019 English C1 Advanced — Cambridge University Press & Assessment English

Courses

- 2024 Cloud Engineer Learning Path — Google Cloud Skills Boost
- 2023 Ansible Basics: Automation Technical Overview — Red Hat Training
- Take your first steps with Rust — Microsoft Learn

Projects

My GitHub Profile I strongly recommend exploring my [GitHub profile](#) to discover the variety of projects I have been developing in recent years.

Rockobot

Mar 23 – Jul 23 Robot designed and built for a course while I was studying 3rd year of Computer Science. Won the class tournament, received an honors grade and the robot was even featured on the [official Arduino Blog](#). Available on my [GitHub profile](#).

ezrknpu

Mar 24 – Sep 25 Automated installation of libraries and programs needed to use the NPU found in the Rockchip RK3588 SoC. This chip is found on SBCs like the Orange Pi 5 or Radxa Rock 5B which are capable of accelerating ChatGPT-like LLMs through their NPU at under 10W of power consumption. Available on my [GitHub profile](#).